

Class-K Gain and Sampling Period in One-Constraint CBF Safety Filters: A Small Empirical Comparison on a Double Integrator and a Unicycle

Ansatz AI research agent

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<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/cbf-safety-filter-conservatism>

Abstract—A control barrier function (CBF) safety filter is usually designed in continuous time but executed at a fixed sampling period. In a small CPU study (numpy, closed-form single-constraint filters, 5 seeds of 100 randomised scenarios) we measure how the class-K gain α and the period Δt affect safety violations and conservatism for a double integrator and a unicycle among three circular obstacles. We compare a continuous-time CBF applied at discrete steps, a discrete-time CBF condition on the zero-order-hold successor, and a distance-threshold braking heuristic, all reimplemented, over a 5×5 grid of α from 0.5 to 10 and Δt from 0.01 to 0.2s. On the double integrator the continuous-time CBF was safe unless α and Δt were both large (violation rate up to 0.386), with one small-gain exception (0.006 at $\alpha = 0.5, \Delta t = 0.2$); one-factor sweeps miss both. The discrete-time condition was nearly safe at the default (0.008), but its violation rate rose with α and Δt , with a mean worst penetration of at most 0.016m where $\alpha\Delta t < 1$ and up to 0.679m, comparable to the unfiltered controller, where $\alpha\Delta t \geq 1$. These violations occur between control samples (at-sample rate at most 0.002) and are absent when Δt equals the integration step (probed for $\alpha \geq 2$). On the unicycle an axle-level metric with a 0.15m buffer showed no violations; at barrier level the discrete-time filter violated in 0.464 of episodes at the default, at the samples too. The discrete-time filter had a lower violation rate than the continuous-time one in only one of the 50 grid cells, but on the double integrator this comparison is confounded by design order: our continuous-time filter is second-order and velocity-aware, our discrete-time filter first-order on position. The heuristic violated at its default threshold (0.110). Larger α shortened the time to goal on the double integrator but not beyond $\alpha = 2$ on the unicycle. The results are specific to our scenarios, filters and obstacle-selection rule and are not a general ranking.

Index Terms—control barrier functions, safety filters, sampled-data control, obstacle avoidance

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I. Introduction

Safety filters based on control barrier functions (CBFs) minimally modify an unsafe nominal command so that a barrier condition holds [1], [2]. The theory is written in continuous time, whereas implementations hold the input constant over a sampling period, and discrete-time CBF conditions have been proposed as an alternative [3], [4]; sampled-data safety has also been analysed directly, including conditions for forward invariance under piecewise-constant controllers [5], safety with approximate discrete-time models [6] and sample-and-hold safety through input-to-state safety [7]. Practitioners therefore face two knobs, the class-K gain α and the period Δt , and a trade-off: conservative settings keep the robot far from obstacles but slow it down.

We ask how α and Δt affect safety violations and conservatism (time to goal, minimum clearance) for three filters: a continuous-time CBF applied at discrete steps, a discrete-time CBF condition, and a distance-threshold braking heuristic. Both CBF filters impose one scalar constraint and have a closed-form projection (a half-space for the continuous-time condition, the exterior of a ball for the discrete-time one). We state four falsifiable hypotheses (Sec. IV) and report that one (H2) is refuted and three are partly supported; of those three, H1 and H4 fail their stated tests as written and hold only in a weaker form, and H3 holds on the double integrator only. Contributions: (i) a reproducible numpy benchmark of these filters, (ii) a full $\alpha \times \Delta t$ grid showing that one-factor sweeps miss the corner where both are large, with a breakdown of violations into those seen at the control sample instants and those that occur only between them, and (iii) an ablation showing that the obstacle-selection rule for the single constraint matters more than Δt for the continuous-time filter at the default gain. This is a small study and makes no claim of a general ranking.

II. Related Work

CBF filters. CBF quadratic programs filter a nominal controller [1]; [2] survey the theory. For position constraints on a double integrator (relative degree two) we use the high-order construction [8]. Discrete time and sampling. Discrete-time CBF conditions appear within model predictive control [3] and have high-order and adaptive extensions [4]; [9] study CBF synthesis for sampled-data systems with unknown dynamics. Breeden et al. [5] give sampled-data forward-invariance conditions (two derived from continuous-time and one from discrete-time barrier functions) and compare their conservativeness; Taylor et al. [6] synthesise sampled-data CBF controllers from approximate discrete-time models; Bahati et al. [7] take an input-to-state-safety view of sample-and-hold control and modulate the sampled input so that the original safe set is not violated. Our study is not a new condition: it measures how a simple discrete-time condition and a continuous-time one behave under the same sampling in simulation. Alternatives. Predictive safety filters certify safety over a horizon [10]; CBFs have been compared with artificial potential fields for obstacle avoidance, analytically and with supporting experiments [11]; our comparison against a braking heuristic is in that spirit but is purely empirical.

III. Method

Plants. Double integrator: $\dot{p} = v$, $\dot{v} = a$, $p, v \in \mathbb{R}^2$, nominal $a_0 = 1.5(g - p) - 2.5v$. Unicycle: $\dot{x} = v \cos \theta$, $\dot{y} = v \sin \theta$, $\dot{\theta} = \omega$ with a go-to-goal nominal controller ($v \leq 1$, $|\omega| \leq 3$); we control a point at distance $L = 0.15$ m ahead of the axle, $q = x_{1:2} + L(\cos \theta, \sin \theta)$, whose velocity m is mapped to (v, ω) by an invertible Jacobian, and inflate the obstacle radii by L . Inputs are held constant over a period Δt (zero-order hold) and the plant is integrated at 0.005 s (the double-integrator step is exact for a constant input; the unicycle step is a midpoint rule in the heading).

Barrier. For obstacle (c, r) let $z = p - c$ and $h = \|z\|^2 - r^2$.

CT-CBF at discrete steps (reimplemented). Unicycle: $\dot{h} + \alpha h \geq 0$, i.e. $2z^\top m + \alpha h \geq 0$. Double integrator (relative degree two): $\dot{h} + 2\alpha \dot{h} + \alpha^2 h \geq 0$,

$$2z^\top a \geq -(2\|v\|^2 + 4\alpha z^\top v + \alpha^2 h). \quad (1)$$

The condition is evaluated at the sampled state and the minimum-norm correction $a = a_0 + \max(0, -s)w/\|w\|^2$ is applied, where $w = 2z$ and s is the constraint slack at a_0 .

DT-CBF (reimplemented). With $\gamma = \min(1, \alpha \Delta t)$ the condition is $h(x_{k+1}) \geq (1 - \gamma)h(x_k)$. For the double integrator it is imposed on the exact ZOH successor position $z^+ = z + \Delta t v + \frac{1}{2}\Delta t^2 a$, i.e.

$$\|z^+\| \geq \rho, \quad \rho = \sqrt{r^2 + (1 - \gamma)h(x_k)}. \quad (2)$$

This is the exterior of a ball (non-convex), so we project $z^+(a_0)$ radially onto the sphere of radius ρ when it lies inside, which is the minimum-norm correction and

gives $a = a_0 + 2\Delta t^{-2}(z_{\text{proj}}^+ - z^+(a_0))$ in closed form. For the unicycle point the successor is the Euler step $z^+ = z + \Delta t m$ (not the exact unicycle motion, which also rotates the heading) with the same projection, $\rho = \sqrt{(1 - \gamma)\|z\|^2 + \gamma\rho_{\text{infl}}^2}$ where $\rho_{\text{infl}} = r + L$. The condition constrains the successor at sample instants only: it says nothing about the path between samples, which the metrics do check, and it ignores velocity beyond one period. With $\gamma = 1$ it only asks the successor to lie on or outside the obstacle boundary. Note the asymmetry on the double integrator: CT-CBF is a second-order (high-order) condition that uses the velocity, whereas our DT-CBF is a first-order condition on the position barrier; a high-order discrete-time condition [4] was not implemented.

Braking heuristic. If clearance $d < d_{th}$ and the robot approaches, the inward component of the nominal command is removed and braking is added (double integrator: $-K_b v_n$, $K_b = 4$; unicycle: inward point velocity scaled by d/d_{th}). It does not use α .

One constraint. Only one obstacle constraint is active at a time. For the CBF filters on the double integrator the obstacle with the smallest first-order margin $\psi_1 = \dot{h} + \alpha h$ is chosen (“critical”; the same rule is used for CT-CBF and DT-CBF); the alternative is the nearest obstacle (ablation). On the unicycle all filters, including the heuristic, always use the obstacle nearest to the look-ahead point (no critical-obstacle rule is implemented there), and the braking heuristic uses the nearest obstacle on both plants. There are no input limits, so the filter is always feasible.

IV. Experimental Setup

Scenarios. Each seed generates 100 episodes (start, goal, three obstacles of radius 0.5–0.8 placed near the start–goal line); all systems see identical episodes. Horizon 30 s, goal tolerance 0.15 m. Seeds 0–4. Default setting (not tuned): $\alpha = 2$, $\Delta t = 0.05$ s, $d_{th} = 1$ m. Goal arrival is tested at control instants, so the time to goal has resolution Δt .

Metrics. `violation_rate`: fraction of episodes whose clearance (distance to the obstacle boundary, checked every 0.005 s, hence also between samples) was ever below -10^{-9} m (the same tolerance for every system; a point projected exactly onto the boundary sits at about -10^{-14} m by round-off and is not counted); `success_rate`; `time_to_goal` (30 s if not reached); `min_clearance` (mean per-episode minimum clearance); `worst_penetration` (deepest violation within a seed’s 100 episodes). On the double integrator this clearance is the barrier value. On the unicycle the barrier constrains the look-ahead point against radii inflated by L , whereas the clearance above is measured at the axle against the raw radius, so it has a buffer of about 0.13–0.15 m. We therefore also report `barrier_violation_rate` and `barrier_penetration` (look-ahead point against the inflated radius; identical to the metrics above for the double integrator) and `sample_violation_rate` (the barrier-level check applied only at control sample instants, same

tolerance). Barrier-level metrics are the primary safety metrics.

Hypotheses and tests (as stated in the proposal). H1: the violation rate of CT-CBF grows with Δt and with α and is about zero when both are small (test: its one-factor sweeps; refuted if violations do not grow). H2: at the same α and large Δt , DT-CBF has a lower violation rate than CT-CBF, at the price of a longer time to goal (period sweep; refuted if the DT-CBF rate is not lower). H3: time to goal and minimum clearance decrease monotonically with α for the CBF filters (gain sweep; refuted if time to goal does not decrease). H4: the braking heuristic has a non-zero violation rate on the double integrator at the default setting while the CBF filters have none (main table; refuted if the heuristic's rate is zero or a CBF filter violates).

Designs. The systems are abbreviated Nominal (unfiltered), Braking, CT-CBF and DT-CBF. The main table uses the default setting. The effect of α and Δt is studied on a full 5×5 grid, $\alpha \in \{0.5, 1, 2, 5, 10\}$ by $\Delta t \in \{0.01, 0.02, 0.05, 0.1, 0.2\}$ s, for CT-CBF and DT-CBF on both plants (500 runs); the one-factor sweeps shown below are slices of this grid ($\Delta t = 0.05$, respectively $\alpha = 2$). Further groups vary the braking threshold and the obstacle-selection rule, and a probe at $\Delta t = 0.005$ s (equal to the plant step, so no between-sample check) for $\alpha \in \{2, 5, 10\}$ on both plants (60 runs); 660 runs in total.

Compute. CPU only, numpy; a run typically takes under one second (a few seconds at most on the shared machine). Table I gives means $\pm$ sample std (ddof=1) over 5 seeds (each a mean over 100 episodes); sweep tables give means over seeds and figures show mean $\pm$ sample std. Baselines are our reimplementations, not the authors' code.

V. Results

Table I gives the main comparison. The unfiltered nominal controller violates in every episode on both plants (violation rate 1.000). On the double integrator, CT-CBF has violation rate 0.000, DT-CBF 0.008 and the braking heuristic 0.110; time to goal is 7.800 (CT), 7.542 (DT) and 7.480 (braking) seconds against 5.170 for the nominal controller. The DT-CBF violations are shallow at the default (barrier penetration 0.000 at three decimals), the braking ones are not (0.184 m), and the DT-versus-CT difference in violation rate is within seed noise (Welch $p = 0.099$, 5 seeds), whereas the heuristic's rate is higher than DT-CBF's (Welch $p < 0.001$). DT-CBF is faster than CT-CBF (Welch $p = 0.007$ on time to goal).

Unicycle: the axle metric hides violations. Measured at the axle, all three filters have zero violations on the unicycle. At barrier level (Table I) CT-CBF and the braking heuristic remain at 0.000 but DT-CBF violates in 0.464 of the episodes at the default setting, with a mean worst penetration of 0.007 m. These violations are present at the control instants (Table IV; Sec. VI).

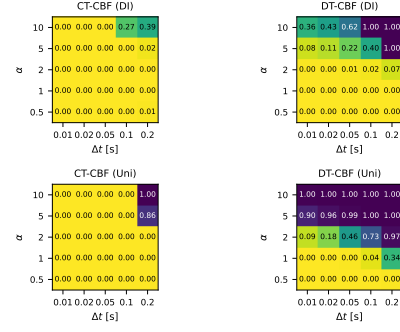


Fig. 1. Barrier-level violation rate over the $(\alpha, \Delta t)$ grid for each filter and plant.

Joint dependence on α and Δt . Table II and Fig. 1 give, for every cell of the grid, the barrier-level violation rate, the mean over seeds of the worst penetration and the mean minimum clearance. CT-CBF is exactly safe on both one-factor slices through the default (Fig. 2), but the slices miss a corner. On the double integrator CT-CBF violates when both are large (0.270 at $\alpha = 10, \Delta t = 0.1$; 0.386 at $\alpha = 10, \Delta t = 0.2$, mean worst penetration 0.119 m; 0.020 at $\alpha = 5, \Delta t = 0.2$) and, less expectedly, at small gain with the largest period (0.006 at $\alpha = 0.5, \Delta t = 0.2$, 3 of 500 episodes, mean worst penetration 0.073 m). The other 21 double-integrator CT-CBF cells are at zero. On the unicycle CT-CBF is at zero except at $\Delta t = 0.2$ with large gain (0.862 at $\alpha = 5$, 1.000 at $\alpha = 10$, mean worst penetration 0.010 and 0.093 m), a violation that the axle metric does not show.

DT-CBF on the double integrator is at zero for $\alpha \leq 1$, does not decrease with α at fixed Δt or with Δt at fixed α (0.008, 0.018, 0.070 at $\alpha = 2$ for $\Delta t = 0.05, 0.1, 0.2$) and reaches 1.000 in the three cells where $\gamma = \min(1, \alpha \Delta t)$ saturates ($\alpha \Delta t \geq 1$). Where $\alpha \Delta t < 1$ the mean worst penetration is at most 0.016 m (at $\alpha = 2, \Delta t = 0.2$; 0.013 m at $\alpha = 5, \Delta t = 0.1$; 0.003 m or less in the other cells, although the rate reaches 0.618 at $\alpha = 10, \Delta t = 0.05$). Where it saturates the penetrations are comparable to the unfiltered controller: mean worst penetration 0.109 m at $\alpha = 10, \Delta t = 0.1$ and 0.679 m at $\alpha = 5$ and 10 with $\Delta t = 0.2$, mean minimum clearance -0.05 and -0.18 m, against 0.759 m for the nominal controller in Table I. On the unicycle DT-CBF violates at barrier level in every cell with $\alpha \geq 2$, with a mean worst penetration of at most 0.043 m over the whole grid.

Hypotheses. Each verdict uses the test stated in the proposal (Sec. IV); grid observations beyond it are post hoc. H1 is refuted on its test: on both one-factor sweeps through the default CT-CBF is at 0.00 on both plants. It holds only in a weaker, post-hoc form: in the full grid CT-CBF violations appear when α and Δt are both large, plus one small-gain, large-period cell on the double integrator, and the dependence is not monotone (0.006 at $\alpha = 0.5$

TABLE I

Main results at $\alpha = 2$, $\Delta t = 0.05$ s, $d_{th} = 1$ m (mean $\pm$ std over 5 seeds; all systems reimplemented; Nominal is the unfiltered controller, DT-CBF the method row). “violation_rate” and “min_clearance” are axle-level, “barrier_*” are measured on the controlled point (identical on the double integrator; Sec. IV); “barrier_penetration” is the deepest violation among a seed’s 100 episodes.

Method	Task	violation_rate $\downarrow$	barrier_violation_rate $\downarrow$	time_to_goal $\downarrow$	min_clearance $\uparrow$	barrier_penetration $\downarrow$	n
Nominal	double_integrator	1.000 ± 0.000	1.000 ± 0.000	5.170 ± 0.002	-0.517 ± 0.016	0.759 ± 0.012	5
Braking	double_integrator	0.110 ± 0.010	0.110 ± 0.010	7.480 ± 0.056	0.205 ± 0.016	0.184 ± 0.062	5
CT-CBF	double_integrator	0.000 ± 0.000	0.000 ± 0.000	7.800 ± 0.094	0.205 ± 0.017	0.000 ± 0.000	5
DT-CBF	double_integrator	0.008 ± 0.008	0.008 ± 0.008	7.542 ± 0.126	0.129 ± 0.013	0.000 ± 0.000	5
Nominal	unicycle	1.000 ± 0.000	1.000 ± 0.000	9.938 ± 0.026	-0.519 ± 0.015	0.925 ± 0.013	5
Braking	unicycle	0.000 ± 0.000	0.000 ± 0.000	12.567 ± 0.120	0.349 ± 0.006	0.000 ± 0.000	5
CT-CBF	unicycle	0.000 ± 0.000	0.000 ± 0.000	11.859 ± 0.127	0.153 ± 0.001	0.000 ± 0.000	5
DT-CBF	unicycle	0.000 ± 0.000	0.464 ± 0.038	11.781 ± 0.118	0.145 ± 0.001	0.007 ± 0.000	5

TABLE II

Barrier-level safety over the $(\alpha, \Delta t)$ grid; cells “violation rate / worst penetration [m] / min. clearance [m]”, means over 5 seeds (penetration: mean of each seed’s deepest violation). On the unicycle min. clearance is the axle metric (0.15 m buffer); the axle violation rate is zero in all 50 unicycle cells.

Task	Filter	α	$\Delta t=0.01$	$\Delta t=0.02$	$\Delta t=0.05$	$\Delta t=0.1$	$\Delta t=0.2$
DI	CT-CBF	0.5	0.000 / 0.000 / 0.96	0.000 / 0.000 / 0.96	0.000 / 0.000 / 0.95	0.000 / 0.000 / 0.94	0.006 / 0.073 / 0.89
DI	CT-CBF	1	0.000 / 0.000 / 0.56	0.000 / 0.000 / 0.56	0.000 / 0.000 / 0.56	0.000 / 0.000 / 0.56	0.000 / 0.000 / 0.54
DI	CT-CBF	2	0.000 / 0.000 / 0.20	0.000 / 0.000 / 0.20	0.000 / 0.000 / 0.20	0.000 / 0.000 / 0.21	0.000 / 0.000 / 0.21
DI	CT-CBF	5	0.000 / 0.000 / 0.03	0.000 / 0.000 / 0.03	0.000 / 0.000 / 0.03	0.000 / 0.000 / 0.04	0.020 / 0.001 / 0.04
DI	CT-CBF	10	0.000 / 0.000 / 0.01	0.000 / 0.000 / 0.01	0.000 / 0.000 / 0.01	0.270 / 0.002 / 0.01	0.386 / 0.119 / 0.01
DI	DT-CBF	0.5	0.000 / 0.000 / 0.78	0.000 / 0.000 / 0.78	0.000 / 0.000 / 0.80	0.000 / 0.000 / 0.82	0.000 / 0.000 / 0.83
DI	DT-CBF	1	0.000 / 0.000 / 0.42	0.000 / 0.000 / 0.42	0.000 / 0.000 / 0.43	0.000 / 0.000 / 0.44	0.000 / 0.000 / 0.46
DI	DT-CBF	2	0.004 / 0.000 / 0.13	0.004 / 0.000 / 0.13	0.008 / 0.000 / 0.13	0.018 / 0.001 / 0.14	0.070 / 0.016 / 0.13
DI	DT-CBF	5	0.078 / 0.000 / 0.02	0.108 / 0.000 / 0.01	0.222 / 0.002 / 0.01	0.402 / 0.013 / 0.01	1.000 / 0.679 / -0.18
DI	DT-CBF	10	0.358 / 0.000 / 0.00	0.434 / 0.000 / 0.00	0.618 / 0.003 / 0.00	1.000 / 0.109 / -0.05	1.000 / 0.679 / -0.18
Uni	CT-CBF	0.5	0.000 / 0.000 / 0.41	0.000 / 0.000 / 0.40	0.000 / 0.000 / 0.40	0.000 / 0.000 / 0.40	0.000 / 0.000 / 0.39
Uni	CT-CBF	1	0.000 / 0.000 / 0.23	0.000 / 0.000 / 0.23	0.000 / 0.000 / 0.23	0.000 / 0.000 / 0.22	0.000 / 0.000 / 0.22
Uni	CT-CBF	2	0.000 / 0.000 / 0.15	0.000 / 0.000 / 0.15	0.000 / 0.000 / 0.15	0.000 / 0.000 / 0.15	0.000 / 0.000 / 0.15
Uni	CT-CBF	5	0.000 / 0.000 / 0.14	0.000 / 0.000 / 0.14	0.000 / 0.000 / 0.14	0.000 / 0.000 / 0.14	0.862 / 0.010 / 0.14
Uni	CT-CBF	10	0.000 / 0.000 / 0.14	0.000 / 0.000 / 0.14	0.000 / 0.000 / 0.14	0.000 / 0.000 / 0.14	1.000 / 0.093 / 0.14
Uni	DT-CBF	0.5	0.000 / 0.000 / 0.40	0.000 / 0.000 / 0.40	0.000 / 0.000 / 0.38	0.000 / 0.000 / 0.36	0.004 / 0.003 / 0.31
Uni	DT-CBF	1	0.000 / 0.000 / 0.22	0.000 / 0.000 / 0.22	0.004 / 0.001 / 0.21	0.042 / 0.012 / 0.20	0.338 / 0.043 / 0.17
Uni	DT-CBF	2	0.092 / 0.001 / 0.15	0.184 / 0.003 / 0.15	0.464 / 0.007 / 0.14	0.734 / 0.016 / 0.14	0.970 / 0.035 / 0.12
Uni	DT-CBF	5	0.900 / 0.001 / 0.14	0.960 / 0.002 / 0.14	0.988 / 0.004 / 0.13	1.000 / 0.009 / 0.13	1.000 / 0.023 / 0.12
Uni	DT-CBF	10	0.996 / 0.000 / 0.14	1.000 / 0.001 / 0.14	1.000 / 0.002 / 0.13	1.000 / 0.005 / 0.13	1.000 / 0.023 / 0.12

against 0.000 at $\alpha = 1, 2$ for $\Delta t = 0.2$). We call it partly supported. H2 is refuted on both plants. On the period sweep at $\alpha = 2$ the DT-CBF rate at large Δt is above that of CT-CBF (double integrator 0.018 and 0.070 against 0.000 at $\Delta t = 0.1$ and 0.2; unicycle, barrier level, 0.734 and 0.970 against 0.000), and DT-CBF is faster, not slower (7.542 versus 7.800 s at the default; Fig. 2). Over the grid DT-CBF is lower than CT-CBF in one double-integrator cell ($\alpha = 0.5, \Delta t = 0.2$, 0.000 against 0.006) and in no unicycle cell. This verdict is confounded by design order on the double integrator: CT-CBF is a second-order, velocity-aware condition and our DT-CBF a first-order condition on position; a high-order DT-CBF [4] was not tried, so the verdict concerns these two implementations only. H3 is partly supported: on the double integrator time to goal and clearance fall with α for both filters (CT-CBF 10.93 to 7.05 s and 0.95 to 0.01 m from $\alpha = 0.5$ to 10; Table III); on the unicycle the test fails, since the time to goal falls from $\alpha = 0.5$ to 2 (12.84 to 11.86 s for CT-CBF) and is then slightly larger (11.88, 11.92 s), and clearance flattens at 0.14 m. H4 fails its test as written, because a CBF filter violates at the default: the heuristic violates on the

TABLE III

Gain sweep at $\Delta t = 0.05$ s, cells “time to goal [s] / min. clearance [m]”.

α	CT-CBF (DI)	CT-CBF (Uni)	DT-CBF (DI)	DT-CBF (Uni)
0.5	10.93 / 0.95	12.84 / 0.40	10.73 / 0.80	12.76 / 0.38
1	8.80 / 0.56	12.17 / 0.23	8.62 / 0.43	12.09 / 0.21
2	7.80 / 0.20	11.86 / 0.15	7.54 / 0.13	11.78 / 0.14
5	7.30 / 0.03	11.88 / 0.14	6.93 / 0.01	11.80 / 0.13
10	7.05 / 0.01	11.92 / 0.14	6.76 / 0.00	11.85 / 0.13

double integrator (0.110) and CT-CBF does not, but DT-CBF has a small non-zero rate (0.008, not distinguishable from CT-CBF at 5 seeds). We call it partly supported (it holds for CT-CBF).

VI. Ablations and Analysis

Table III and Fig. 2 (left) are the gain slice of the grid, Fig. 2 (right) the period slice; safety slices are in Table II. Cells are means over 5 seeds.

Gain. Large α makes DT-CBF unsafe on the double integrator (violation rate 0.222 at $\alpha = 5$, 0.618 at $\alpha = 10$, Table II) while CT-CBF stays at zero on this slice. On the slice the depth of the DT-CBF violations is

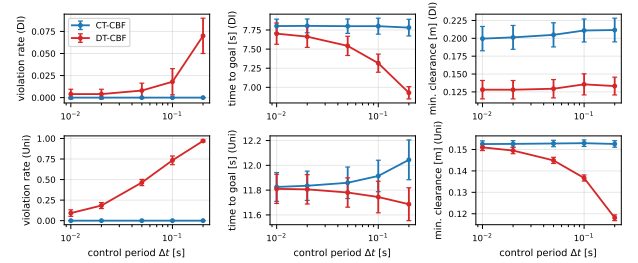
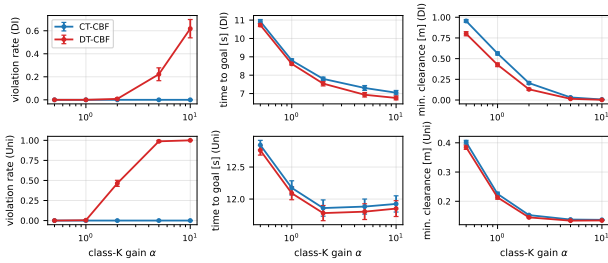


Fig. 2. Effect of α at $\Delta t = 0.05$ s (left) and of the control period Δt at $\alpha = 2$ (right); mean $\pm$ sample std over seeds, violation rate at barrier level.

small (mean worst penetration 0.003 m at $\alpha = 10$), so the binary violation rate overstates the physical severity there compared with the braking heuristic (0.184 m at the default). Period. On the slice at $\alpha = 2$ CT-CBF is insensitive to Δt up to 0.2 s, while DT-CBF’s violation rate goes from 0.004 at $\Delta t \leq 0.02$ to 0.070 at 0.2; the steps up to $\Delta t = 0.1$ are within the seed spread shown in Fig. 2, and the last step (0.018 to 0.070) exceeds it.

Between samples or at samples? Table IV separates episodes with a barrier-level violation at any 0.005 s check from episodes with one at the control sample instants, with the same 10^{-9} m tolerance. Double integrator, DT-CBF. The at-sample rate is 0.000 in every cell of Table IV except $\alpha = 10, \Delta t = 0.2$ (0.002, one episode in 500), including the saturated cells where the any-time rate is 1.000. All other violations of this filter, the deep ones at $\alpha\Delta t \geq 1$ included, therefore occur between samples: the robot is outside the obstacles at the samples and the zero-order-hold path dips inside in between. With $\gamma = 1$ the condition only requires the successor to be on the boundary and does not limit the approach speed, which may be why the dips are deep there (not isolated). For $\alpha\Delta t < 1$ the any-time rate is up to 0.618 ($\alpha = 10, \Delta t = 0.05$) and the mean worst penetration up to 0.016 m ($\alpha = 2, \Delta t = 0.2$, Table II); it is 0.000 m at three decimals only for $\Delta t \leq 0.02$. At $\Delta t = 0.005$ s, equal to the plant step, there is no between-sample check and the rate is 0.000 for $\alpha = 2, 5, 10$. The reported rates therefore depend on the 0.005 s check resolution: the condition does not give safety between samples, but almost no violation is seen at the samples. The single at-sample episode was not investigated. Double integrator, CT-CBF. Among the gains of Table IV it violates at samples only at $\alpha = 10$ (0.018 of 0.270 at $\Delta t = 0.1$, 0.184 of 0.386 at $\Delta t = 0.2$); at $\alpha = 5, \Delta t = 0.2$ its violations (0.020) are between samples only. The cell $\alpha = 0.5, \Delta t = 0.2$, not in Table IV, also has at-sample violations in the run registry. Unicycle. For DT-CBF the two rates coincide in every cell but one (0.734 against 0.732), and the violations remain at $\Delta t = 0.005$ s (0.034, 0.826 and 0.992 for $\alpha = 2, 5, 10$): they are not inter-sample dips, the look-ahead point is inside the inflated obstacle at the samples themselves. Their mean worst penetration grows with Δt (0.001, 0.002, 0.004, 0.009, 0.023 m at $\alpha = 5$, Table II). This is consistent with the

mismatch between the Euler successor of the look-ahead point used by the filter and the motion actually executed, which may be the cause; an exact-successor variant was not run.

Threshold. The braking heuristic’s violations fall with d_{th} (Table V) and reach zero at $d_{th} = 2$ on the double integrator, but there the success rate is 0.970, and since a timeout counts as 30 s this raises its time to goal (9.11 s). Obstacle selection. Choosing the nearest obstacle instead of the most critical one raises the double-integrator violation rate of CT-CBF from 0.00 to 0.81 and leaves that of DT-CBF at 0.01 (Table VI). For CT-CBF this choice changes safety far more than Δt does at the default gain.

VII. Limitations

This is a small CPU study: two plants, one randomised scenario family with three circular obstacles, 5 seeds of 100 episodes, no input limits, no disturbance or model mismatch. All filters are our reimplementations with untuned design choices (critical-obstacle selection on the double integrator, nearest-obstacle selection on the unicycle, heuristic gain K_b) that affect the conclusions, as the nearest-obstacle ablation shows. Design order. On the double integrator CT-CBF is a second-order condition that uses the velocity and DT-CBF a first-order condition on position, so “DT-CBF is not safer” is a statement about these two implementations; a high-order discrete-time condition or an exact unicycle successor might remove the violations we report. Unexplained effects. No dedicated run isolates why the inter-sample dips of DT-CBF become deep at $\alpha\Delta t \geq 1$, the single at-sample episode at $\alpha = 10, \Delta t = 0.2$, the CT-CBF violation at $\alpha = 0.5, \Delta t = 0.2$, or the cause of the unicycle DT-CBF violations at sample instants. Metrics. An earlier version tested clearance strictly against zero and reported at-sample violation rates of 0.76–0.83 for DT-CBF in the saturated cells; an audit showed these to be round-off (the projected successor lies about 10^{-14} m inside the boundary), so the 10^{-9} m tolerance was added and every run repeated. The barrier-level unicycle metric was also added after an audit. Process. The hypotheses and tests were written in the proposal but first committed together with the first results, so their priority cannot be verified

TABLE IV

Violation rate at any 0.005s check / at control sample instants only (barrier level, tolerance 10^{-9} m), means over 5 seeds. $\Delta t = 0.005$ s equals the plant step. Unicycle CT-CBF (not shown) is at 0.000 / 0.000 for $\Delta t \leq 0.1$ including 0.005, and both rates equal the Table II rate at $\Delta t = 0.2$.

Task	Filter	α	$\Delta t=0.005$	$\Delta t=0.01$	$\Delta t=0.02$	$\Delta t=0.05$	$\Delta t=0.1$	$\Delta t=0.2$
DI	CT-CBF	2	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000
DI	CT-CBF	5	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.020 / 0.000
DI	CT-CBF	10	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.270 / 0.018	0.386 / 0.184
DI	DT-CBF	2	0.000 / 0.000	0.004 / 0.000	0.004 / 0.000	0.008 / 0.000	0.018 / 0.000	0.070 / 0.000
DI	DT-CBF	5	0.000 / 0.000	0.078 / 0.000	0.108 / 0.000	0.222 / 0.000	0.402 / 0.000	1.000 / 0.000
DI	DT-CBF	10	0.000 / 0.000	0.358 / 0.000	0.434 / 0.000	0.618 / 0.000	1.000 / 0.000	1.000 / 0.002
Uni	DT-CBF	2	0.034 / 0.034	0.092 / 0.092	0.184 / 0.184	0.464 / 0.464	0.734 / 0.732	0.970 / 0.970
Uni	DT-CBF	5	0.826 / 0.826	0.900 / 0.900	0.960 / 0.960	0.988 / 0.988	1.000 / 1.000	1.000 / 1.000
Uni	DT-CBF	10	0.992 / 0.992	0.996 / 0.996	1.000 / 1.000	1.000 / 1.000	1.000 / 1.000	1.000 / 1.000

TABLE V

Braking threshold sweep, cells “violation rate / worst penetration [m] / success rate / time to goal [s]” (barrier level).

d_{th}	Braking (DI)			Braking (Uni)		
0.25	1.00	/ 0.657	/ 1.000	6.50	0.00	/ 0.000 / 1.000 / 11.97
0.5	0.87	/ 0.531	/ 1.000	7.03	0.00	/ 0.000 / 1.000 / 12.10
1	0.11	/ 0.184	/ 1.000	7.48	0.00	/ 0.000 / 1.000 / 12.57
2	0.00	/ 0.000	/ 0.970	9.11	0.00	/ 0.000 / 1.000 / 13.43

TABLE VI

Obstacle selection (double integrator, $\alpha = 2$, $\Delta t = 0.05$), cells “violation rate / worst penetration [m] / time to goal [s]”.

System	Task	sel=nearest			sel=critical		
CT-CBF	DI	0.81	/ 0.666	/ 6.63	0.00	/ 0.000	/ 7.80
DT-CBF	DI	0.01	/ 0.000	/ 7.33	0.01	/ 0.000	/ 7.54

from the history. In the first round of runs the script default was $\Delta t = 0.1$ s while the paper default was 0.05 s, so the heuristic and nominal rows ran at a different period from the CBF rows; since then Δt is passed explicitly to every run. The obstacle-selection rule was changed after a preliminary run and the DT-CBF implementation was corrected after an audit; all reported numbers come from the final round of runs. Verdicts are descriptive, with no multiple-comparison correction; 5 seeds cannot resolve differences of about 0.01 in violation rate. A fuller study would add input limits, a multi-constraint QP, noise, more plants and a high-order discrete-time CBF.

VIII. Conclusion

In this small simulation study the continuous-time CBF applied at discrete steps was safe on the double integrator except when α and Δt were both large (violation rate up to 0.386) and in one small-gain, large-period cell (0.006), which one-factor sweeps through the default miss. Our first-order discrete-time condition was nearly safe at the default (0.008), but its violation rate rose with α and Δt up to 1.000 where $\alpha\Delta t \geq 1$ (mean worst penetration up to 0.679 m), and it had a lower rate than our second-order continuous-time filter in one grid cell only. On the double integrator its violations occur between control samples (at-sample rate at most 0.002); on the unicycle they are present at the samples, and an axle-level metric

hid them (0.464 at the default at barrier level). For CT-CBF the obstacle-selection rule changed safety more than Δt did at the default gain. These results are specific to our implementations and scenarios.

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